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Estimation of longitudinal brain deformations based on stationary velocity fields

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Context

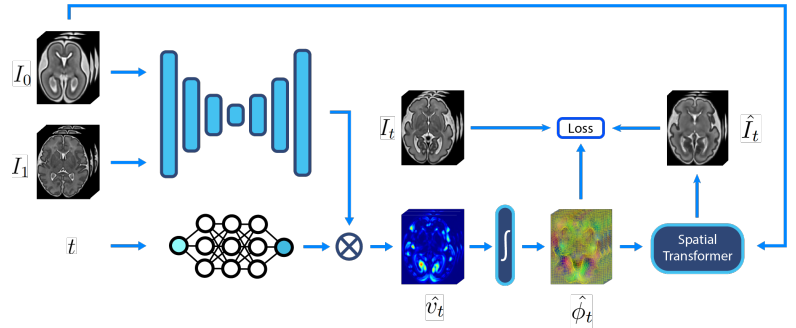
Fetal brain development undergoes rapid and non-linear changes, particularly in cortical folding. A precise understanding of this growth requires estimating the developmental trajectory by capturing the dynamics of structural deformations.

Objective

Estimate the longitudinal brain deformation trajectory by calculating a continuous deformation field $\phi(t)$ from a stationary velocity field (SVF) v , integrating longitudinal fetal MRI data to capture dynamic structural changes over time.

Method

- **SVF Framework:** Predict a SVF between a source image I_0 to a target image I_t by a neural network. The velocity field is integrated to obtain a deformation field, $\hat{\phi}$ then warp the source image with $\hat{\phi}$ to generate a predicted image.
- **SVF Weighting:** SVF weighting by a linear or MLP-based scalar factor t to compute adapted data-driven velocity field $\hat{v}_t$ at any given time step.
- **Optimization:** Data-driven, self-supervised parameterization is achieved by minimizing the error between the predicted image $\hat{I}_t$ and the ground truth I_t for all $t \in \mathcal{T}$ in the longitudinal MRI atlas. Furthermore, the inverse transformation ϕ^{-1} applied to warp I_t to I_0 , as well as a regularization term on the deformation field, are used to enhance smoothness while a segmentation loss enforces accurate structural deformations.



$$\theta^* = \arg \min_{\theta} \sum_{t \in \mathcal{T}} \mathcal{L}_{sim}(\hat{I}_t(\theta), I_t) + \mathcal{L}_{sim}(\hat{I}_0(\theta), I_0) + \alpha [\mathcal{L}_{seg}(\hat{S}_t(\theta), S_t) + \mathcal{L}_{seg}(\hat{S}_0(\theta), S_0)] + \beta [\mathcal{L}_{reg}(\hat{\phi}_t) + \mathcal{L}_{reg}(\hat{\phi}_t^{-1})]$$

Results

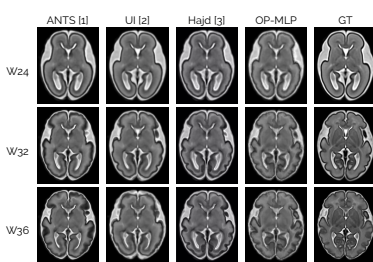


Figure 1. Visual comparison of image deformations obtained using different methods during registration, on the dHCP Atlas, between the week 21 image and the images from weeks 24, 32, and 36, respectively (from top to bottom).

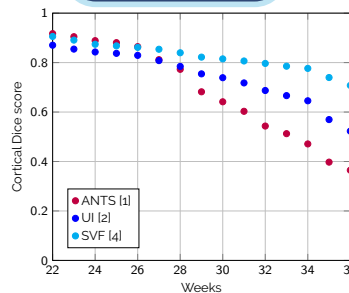
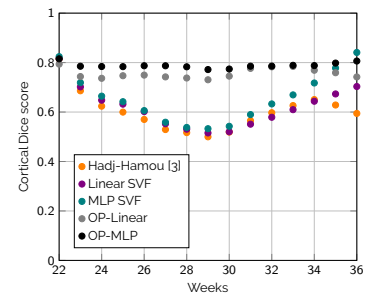


Figure 2. Evolution of the cortical Dice score obtained by pairwise registration (left) and longitudinal methods (right) between the MRI at W-21 and the other images in the dHCP Atlas



Pairwise SVF framework registers the week 21 source image with the week 36 target image, with a maximum Dice cortical score of 70.76%. The optimized linear/MLP estimates a temporal trajectory by optimizing on longitudinal images, maintaining stable cortical Dice coefficients around 80%.

Conclusion

- Data-driven optimized methods based on longitudinal images effectively model the fetal brain deformation trajectory by continuously representing the deformation field defined by a stationary velocity field.
- Non-linear methods achieve slightly better results than the linear approach in capturing the non-linear development trajectory.

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